

Software Engineering (UE24CS341A)

TEAM NUMBER: 007

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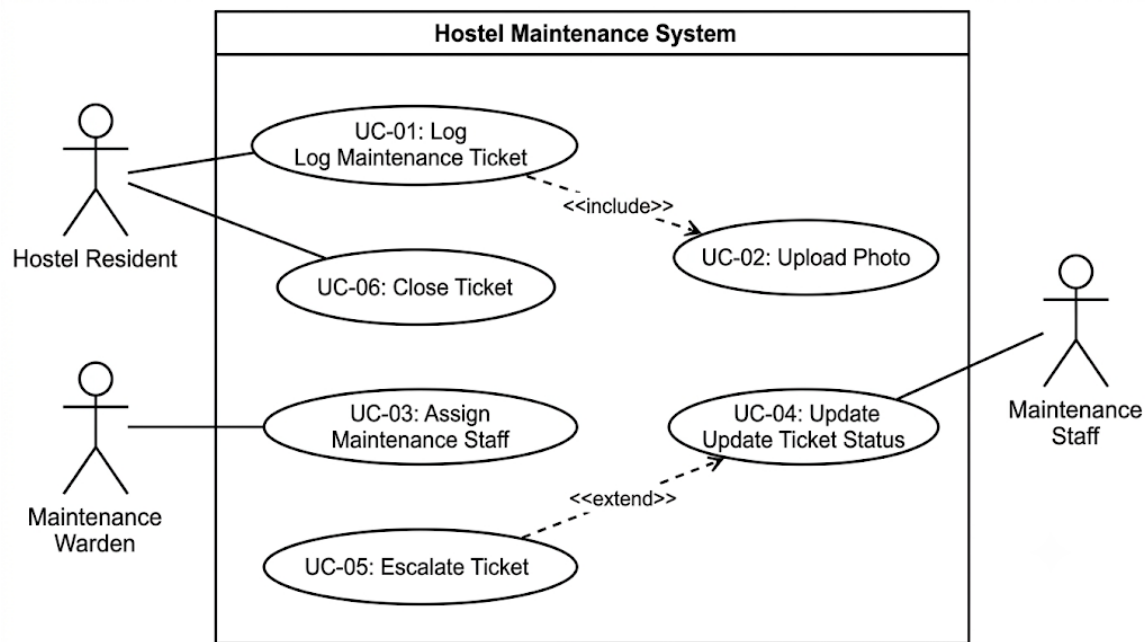
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Lab 1: Hostel Maintenance & Issue Ticketing System

1. Requirements Table

Req ID	Type	Description	Priorit	Acceptance	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue; Fail: Incomplete form accepted without room tag.	Core functionality for issue reporting
FR-002	Functional	The system shall allow the Maintenance Warden to assign an open ticket to a specific Maintenance Staff member.	High	Pass: Warden can select staff from a dropdown and ticket status changes to Assigned.	Necessary for dispatching staff
FR-003	Functional	The system shall allow Maintenance Staff to update the status of an assigned ticket (e.g., In Progress, Resolved).	High	Pass: Staff updates status to Resolved and timestamp is recorded.	Tracks progress of the issue
FR-004	Functional	The system shall require the Hostel Resident to digitally sign off to close a resolved ticket.	Medium	Pass: Ticket status changes to Closed after resident confirms.	Ensures resident satisfaction before closure
FR-005	Functional	The system shall display a dashboard of all active tickets for the Maintenance Warden, sortable by severity and time.	Medium	Pass: Warden sees a list of tickets sortable by severity.	Allows warden to oversee and prioritize work
NFR-001	Non-functional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures urgent issues are not ignored
NFR-002	Non-functional	The system shall have an uptime of 99.9% during the academic semester.	Medium	Pass: Uptime logs show <= 43 minutes of downtime per month.	Residents must be able to report urgent issues

2. UML Use-Case Diagram



3. Actors & Use Cases

Actors: Hostel Resident, Maintenance Warden, Maintenance Staff

Use-Case Flow Specification

Use Case: UC-04 Update Ticket Status

Primary Actor: Maintenance Staff

Preconditions:

1. The Maintenance Staff is authenticated and logged into the system.
2. A maintenance ticket has been raised by a Hostel Resident.
3. The ticket is currently in an active status such as **Open**, **In Progress**, or **Pending**.

Postconditions:

The maintenance ticket status is updated to the selected status, and the updated status is recorded in the system.

Main Success Scenario:

1. Maintenance Staff navigates to the "**Maintenance Tickets**" section.

2. System displays the list of maintenance tickets assigned to or accessible by the Maintenance Staff.
3. Maintenance Staff selects the ticket whose status they want to update.
4. System displays the ticket details, including the current status and maintenance information.
5. Maintenance Staff selects the option to **update the ticket status**.
6. System displays the available ticket statuses.
7. Maintenance Staff selects the appropriate new status.
8. System updates the ticket status and records the date and time of the status change.
9. System displays a success message confirming that the ticket status has been updated.

Alternate Flow:

6a. Invalid status update

1. Maintenance Staff selects an invalid or unavailable status.
2. System displays an error message indicating that the selected status cannot be applied.
3. Maintenance Staff selects a valid status.
4. System updates the ticket status accordingly.
5. Use case continues from Step 8 of the Main Success Scenario.

7a. Maintenance Staff cancels the update

1. Maintenance Staff cancels the status update without selecting a new status.
2. System retains the existing ticket status.
3. Use case ends without making any changes to the ticket.